Light

GCSE Computer Science — Character Sets (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What does the term 'bit' stand for?

- **A.** Binary digit
- **B.** Binary information transfer
- **C.** Basic integer type
- **D.** Byte input terminal

Q2 (1 mark)

How many different characters can be represented using exactly 8 bits?

- **A.** 8 characters

- B. 64 characters
- C. 128 characters
- D. 256 characters

Q3 (1 mark)

In ASCII, the character 'C' has code 67. Using the logical ordering principle, what code does 'B' have?

- A. 66
- B. 65
- C. 68
- D. 78

Q4 (1 mark)

In ASCII, the digit character '0' has code 48. What code does the digit character '5' have?

- A. 5
- B. 50
- C. 53
- D. 55

Q5 (1 mark)

ASCII includes 33 'control characters' (codes 0-31 and 127). What is the purpose of these characters?

- ○ **A.** To display special visual symbols on screen
- ○ **B.** To control devices and format text, such as starting a new line or creating a tab space
- ○ **C.** To represent mathematical operators
- ○ **D.** To store encrypted passwords

Q6 (1 mark)

What is 'Extended ASCII'?

- ○ **A.** A character set that uses 16 bits per character
- ○ **B.** An 8-bit character set with 256 characters, adding extra symbols and accented letters to the original 128-character ASCII
- ○ **C.** A compressed version of ASCII that uses fewer bits
- ○ **D.** A character set exclusively for programming languages

Q7 (1 mark)

Why are character codes often displayed in hexadecimal rather than decimal or binary?

- ○ **A.** Hexadecimal is more compact than binary while still showing the underlying bit structure clearly
- ○ **B.** Hexadecimal uses less electricity than binary
- ○ **C.** Hexadecimal is the only format computers can process

- ○ **D.** Hexadecimal makes characters appear in alphabetical order

Q8 (1 mark)

In Unicode documentation, characters are often written as 'U+0041'. What does this notation represent?

- ○ **A.** A Unicode file format version number
- ○ **B.** A code point - the hexadecimal number 0041 (decimal 65) identifies the character 'A'
- ○ **C.** The file size of a Unicode document
- ○ **D.** The number of bits used to store the character

Q9 (1 mark)

In ASCII, what is the decimal code for the space character?

- ○ **A.** 0
- ○ **B.** 20
- ○ **C.** 32
- ○ **D.** 64

Q10 (1 mark)

Why don't the digit characters '0' to '9' have ASCII codes 0 to 9?

- ○ **A.** Because codes 0-9 don't exist in ASCII
- ○ **B.** Because codes 0-9 are reserved for control characters that perform special functions
- ○ **C.** Because digit characters require 16-bit codes
- ○ **D.** Because ASCII was designed before numbers were invented

Q11 (1 mark)

What is the difference between a character encoding and a font?

- ○ **A.** They are the same thing with different names
- ○ **B.** Character encoding defines which binary codes represent which characters; fonts define how those characters appear visually
- ○ **C.** Fonts store characters in binary; encodings make them look pretty
- ○ **D.** Character encoding only works with Unicode; fonts work with ASCII

Q12 (1 mark)

What is a Byte Order Mark (BOM) in Unicode files?

- ○ **A.** A special character sequence at the start of a file that identifies the encoding and byte order
- ○ **B.** A virus that corrupts text files
- ○ **C.** The total number of bytes in a file

- ○ **D.** A trademark symbol required in all documents

Q13 (2 marks)

A plain text file contains 750 characters stored using ASCII (8 bits per character). Calculate the file size in bytes.

- ○ **A.** 93.75 bytes
- ○ **B.** 750 bytes
- ○ **C.** 6,000 bytes
- ○ **D.** 6,000 bits

Q14 (2 marks)

A device manufacturer needs to create a character set that can represent exactly 100 different characters. What is the minimum number of bits required per character?

- ○ **A.** 6 bits
- ○ **B.** 10 bits
- ○ **C.** 8 bits
- ○ **D.** 7 bits

Q15 (2 marks)

In ASCII, the character 'P' has code 80. A programmer needs to work out the code for 'K'. Explain the calculation.

- ○ **A.** Add 5 to 80 because K is 5 letters after P
- ○ **B.** Multiply 80 by the position of K in the alphabet
- ○ **C.** Subtract 5 from 80 because K is 5 letters before P in the alphabet, giving 75
- ○ **D.** Divide 80 by 2 because K is halfway through the alphabet

Q16 (2 marks)

A German company creates documents containing umlauts (ä, ö, ü) and the ß character. Compare ASCII and Unicode for this use case.

- ○ **A.** ASCII is better because German is a European language like English
- ○ **B.** Unicode cannot represent European characters, only Asian scripts
- ○ **C.** Standard ASCII cannot reliably represent German special characters; Unicode fully supports them with dedicated code points
- ○ **D.** Both encodings handle German characters identically

Q17 (2 marks)

In ASCII, the uppercase letter 'E' has decimal code 69. What is this as an 8-bit binary number?

- ○ **A.** 01000101
- ○ **B.** 01000100
- ○ **C.** 01100101

- ○ **D.** 00110101

Q18 (2 marks)

Web pages often include a declaration like `<meta charset="UTF-8">`. Why is this important?

- ○ **A.** It makes the web page load faster
- ○ **B.** It determines the colour scheme of the web page
- ○ **C.** It encrypts the web page content for security
- ○ **D.** It tells the browser which character encoding the page uses, ensuring text displays correctly

Q19 (2 marks)

A message is encoded as the ASCII sequence: 67, 65, 84. Using the knowledge that 'A' = 65, decode the message.

- ○ **A.** BAT
- ○ **B.** CAT
- ○ **C.** RAT
- ○ **D.** HAT

Q20 (2 marks)

Calculate the file size for 5,000 characters stored in: (a) ASCII (8 bits), and (b) UTF-32 (32 bits per character). What is the difference?

- ○ **A.** ASCII: 5,000 bytes, UTF-32: 20,000 bytes; difference is 15,000 bytes
- ○ **B.** ASCII: 5,000 bytes, UTF-32: 10,000 bytes; difference is 5,000 bytes
- ○ **C.** ASCII: 625 bytes, UTF-32: 2,500 bytes; difference is 1,875 bytes
- ○ **D.** Both are 5,000 bytes; no difference

Q21 (2 marks)

A software company needs to support French, German, and Spanish text. Compare Extended ASCII and Unicode for this requirement.

- ○ **A.** Extended ASCII is always better for European languages
- ○ **B.** Unicode cannot represent characters with accents
- ○ **C.** Extended ASCII code pages can handle these languages but vary between systems; Unicode provides universal, consistent support
- ○ **D.** Neither encoding supports European accented characters

Q22 (2 marks)

In ASCII, uppercase 'Z' has code 90. Given that there is a consistent gap between uppercase and lowercase letters, what is the code for lowercase 'z'?

- ○ **A.** 91
- ○ **B.** 122
- ○ **C.** 116

○ **D.** 100

Q23 (2 marks)

An email contains special characters but arrives with some characters displaying as '?' or '□'. Explain the likely cause.

○ **A.** The email server automatically removes all special characters for security

○ **B.** The sender's encoding and receiver's encoding don't match, or the receiver's system lacks fonts for those characters

○ **C.** Email systems cannot transmit any characters except basic letters

○ **D.** The '?' and '□' are standard replacements that should be manually translated

Q24 (3 marks)

An e-commerce company operates globally and must store customer names from any country. Their database currently uses ASCII. Analyse the encoding requirements for handling names like 'François', '田中', and 'Мария'.

○ **A.** ASCII is sufficient because all names can be transliterated to English letters

○ **B.** Extended ASCII would work since it includes all world languages

○ **C.** Unicode (UTF-8 or UTF-16) is essential because ASCII cannot represent French accented characters, Chinese characters, or Cyrillic script; each requires code points beyond ASCII's 128-character limit

○ **D.** A custom character set should be created specifically for this company

Q25 (3 marks)

A text editor allows users to convert documents between different character encodings. A user converts a UTF-8 file containing Greek text to ASCII.
Analyse what happens to the Greek characters during this conversion.

○ **A.** Greek characters are automatically translated to their English equivalents

○ **B.** Greek characters cannot be represented in ASCII and will be lost, replaced with '?' or similar substitution characters, or the conversion will fail with an error

○ **C.** Greek characters are compressed to fit within ASCII's available codes

○ **D.** The file size doubles because Greek characters need more space in ASCII

Q26 (3 marks)

A social media platform processes 1 million posts per day, averaging 50 characters each. Compare the daily storage requirements for:

- ASCII (8 bits per character)
- UTF-16 (16 bits per character)

Analyse the storage implications.

○ **A.** ASCII: 50 MB, UTF-16: 100 MB; the 50 MB difference is negligible for modern systems

○ **B.** ASCII: 50 MB, UTF-16: 50 MB; both encodings use identical storage

○ **C.** ASCII: 400 MB, UTF-16: 800 MB; the doubling represents significant cost over time but UTF-16 may be necessary for international content

○ **D.** Storage calculations are irrelevant because text compression eliminates encoding differences

Q27 (3 marks)

A school's student database stores names in ASCII. The school now accepts international students with names like 'Björk' (Icelandic) and 'Nguyễn' (Vietnamese).
Analyse the challenges and benefits of migrating to Unicode.

○ **A.** No migration is needed; students should change their names to ASCII-compatible versions

○ **B.** Migration is straightforward with no challenges since Unicode is backwards-compatible with ASCII

○ **C.** Migration requires careful planning: existing ASCII data transfers safely due to Unicode compatibility, but new storage requirements, software updates, and potential sorting/searching changes must be addressed; the benefit is accurate representation of all student names

○ **D.** The database should use images of names instead of text encoding

Q28 (3 marks)

Programming languages provide functions like `chr()` (converts code to character) and `ord()` (converts character to code).
Analyse why these functions are useful for working with character sets.

○ **A.** They are obsolete functions only used in legacy systems

○ **B.** They only work with Unicode, not ASCII

○ **C.** They allow programmers to work directly with character codes for tasks like encryption, data validation, case conversion, and generating character sequences; understanding the numerical nature of characters enables powerful text manipulation

○ **D.** They are only used for displaying error messages

Q29 (3 marks)

An international company's email system must reliably deliver messages between users worldwide, including content in English, Chinese, Arabic, and Russian.
Analyse the character encoding considerations for this system.

○ **A.** Use ASCII and require all employees to write in English only

○ **B.** Create separate email systems for each language region

○ **C.** Character encoding is handled automatically by email and requires no consideration

○ **D.** Implement UTF-8 as the standard encoding: it handles all languages within a single system, is backwards-compatible with ASCII for English content efficiency, is the standard for email protocols (MIME), and eliminates encoding mismatches that cause garbled text

Q30 (3 marks)

A news website serves content that is 80% English and 20% international languages (including Chinese and Arabic). Compare the suitability of ASCII, UTF-8, and UTF-16 for this use case.

○ **A.** ASCII is best because 80% of content is English, and international content can be transliterated

○ **B.** UTF-16 is best because it handles international characters most efficiently

○ **C.** UTF-8 is optimal: English content (80%) uses 1 byte per character like ASCII, international content uses 2-4 bytes as needed; this provides the best balance of storage efficiency and universal language support for this content mix

○ **D.** All three encodings would produce identical file sizes for this content

Total: **55** marks • Time: **30 min**

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